

The Continuous Colonel Blotto Game

A First-Principles Derivation of the Value, Nash Equilibrium, and Optimal-Strategy Construction

Roadmap. Starting from the most primitive game model, we progressively compress Colonel Blotto into a single maximin object $\underline{v} = \max_P \min_Q U_A(P, Q)$, then **without presupposing any solution** derive: (1) the game has a value; (2) that value equals $n/2$; (3) a symmetric strategy P^* attains it; (4) (P^*, P^*) is a Nash equilibrium; (5) P^* exists as a legitimate joint distribution (explicit construction). The guiding methodological principle throughout: **define the optimization object first, then let the solution grow out of it—never reverse-engineer**. The optimal marginal $F(t) = \frac{nt}{2X}$ is not an assumption but a **conclusion**.

1. Modeling: From Primitive Rules to the Maximin Object

1.1 Formal definitions

Definition 1 (Battlefields). There are $n \geq 2$ battlefields $B = \{1, \dots, n\}$, each of equal value $v_i = 1$.

Definition 2 (Budget). Both players have identical resource budget $X > 0$.

Definition 3 (Pure strategy). An allocation is a vector $x = (x_1, \dots, x_n)$ with $x_i \geq 0$ and $\sum_i x_i = X$. The pure-strategy space is

$$S_X = \left\{ x \in \mathbb{R}_{\geq 0}^n : \sum_{i=1}^n x_i = X \right\},$$

a scaled simplex—closed and bounded, hence compact.

Definition 4 (Per-battlefield payoff).

$$u_i(x_i, y_i) = \begin{cases} 1, & x_i > y_i, \\ \frac{1}{2}, & x_i = y_i, \\ 0, & x_i < y_i. \end{cases}$$

The $\frac{1}{2}$ tie-break is symmetric; we will see it "vanishes by measure zero" under continuous strategies, but it is indispensable at the modeling level.

Definition 5 (Total payoff). $u(x, y) = \sum_{i=1}^n u_i(x_i, y_i)$.

Definition 6 (Mixed strategy & expected payoff). A mixed strategy is a probability measure $P \in \Delta(S_X)$. The two players randomize **independently**, $x \sim P$, $y \sim Q$, with joint law the product measure $P \otimes Q$:

$$U_A(P, Q) = \mathbb{E}_{x \sim P, y \sim Q}[u(x, y)] = \iint u(x, y) dP(x) dQ(y).$$

1.2 Two levels of independence

- **Between players:** $x \perp y$. Both randomize simultaneously and independently; neither sees the other's realization. This is what makes the bilinear integral form valid. This is **not** a Stackelberg game—there is no "deploy after seeing the opponent deploy."
- **Within a player:** $x_i \not\perp x_j$. Locked by the budget $\sum_i x_i = X$ —knowing the first $n - 1$ coordinates fixes the last. This inter-coordinate dependence is the entire source of Colonel Blotto's difficulty.

The $\min_Q$ in maximin is **analytical ordering** (fix P , then sweep Q), not **temporal ordering**; both still commit simultaneously. Analytical ordering $\neq$ temporal ordering.

1.3 Why it collapses to maximin: constant-sum structure

Proposition 1 (Constant-sum). For all x, y , $u(x, y) + u(y, x) = n$.

Per battlefield $1 + 0 = \frac{1}{2} + \frac{1}{2} = 0 + 1 = 1$, summed over n . ■

Lifting to mixed strategies: $U_B(Q, P) = n - U_A(P, Q)$. Hence B rationally does $\max_Q U_B = \max_Q (n - U_A) = \min_Q U_A$.

The opponent is a minimizer. This is not a "worst-case analysis" preference but is **forced** by the zero-sum structure: the opponent's optimum is exactly my worst.

Thus A's security value is

$$\underline{v} = \max_{P \in \Delta(S_X)} \min_{Q \in \Delta(S_X)} U_A(P, Q),$$

and dually $\bar{v} = \min_Q \max_P U_A(P, Q)$.

Lemma 1 (Weak duality). $\underline{v} \leq \bar{v}$.

For any fixed P, Q : $\min_{Q'} U_A(P, Q') \leq U_A(P, Q) \leq \max_{P'} U_A(P', Q)$. The left depends only on P , the right only on Q , holding for all P, Q ; hence $\underline{v} \leq \bar{v}$. Intuition: "the later mover can adapt." ■

2. Expanding the Expectation into Marginals

2.1 Additive structure lowers dimension

By linearity of expectation, $U_A(P, Q) = \sum_{i=1}^n \mathbb{E}[u_i(x_i, y_i)]$. Note we split the **expectation**, not the **distribution**—coordinates remain budget-coupled. Grounding via the definition (expectation of an indicator = probability of the event, $x_i \perp y_i$):

$$\mathbb{E}[u_i(x_i, y_i)] = \Pr(x_i > y_i) + \frac{1}{2}\Pr(x_i = y_i).$$

The gift of additivity. Each single-battlefield expectation touches only the **two-dimensional** joint of the pair (x_i, y_i) , which equals the product of the two marginals (inter-player independence). Therefore the within-player coordinate dependence is **irrelevant to computing** U_A —additivity reduces an n -dimensional joint problem to n two-dimensional problems needing only marginals.

Writing marginal CDFs $F_i^P(t) = \Pr_{x \sim P}(x_i \leq t)$, $F_i^Q(t) = \Pr_{y \sim Q}(y_i \leq t)$, with infinitesimal notation ($\delta t \rightarrow 0$):

$$U_A(P, Q) = \sum_{i=1}^n \sum_t F_i^Q(t) [F_i^P(t + \delta t) - F_i^P(t)] + \underbrace{\frac{1}{2} \sum_i \sum_t \Delta F_i^P(t) \Delta F_i^Q(t)}_{\text{tie term}}.$$

2.2 The tie term vanishes under continuous strategies

Lemma 2 (Ties have measure zero). If the marginal F_i^P is atomless (continuous, no jumps), the tie term is 0.

The single-point probability equals the CDF jump: $\Pr(x_i = t) = F(t) - F(t^-)$. Continuity $\implies F(t^-) = F(t)$ everywhere $\implies \Pr(x_i = t) = 0$. Each tie summand thus carries a factor 0; the sum vanishes. ■

Verification via δt . The probability that " x_i, y_i land in the same δt bin" is $\sum_t f^P f^Q (\delta t)^2$; with $\sim 1/\delta t$ bins the total is $\sim \delta t \rightarrow 0$. The tie term is **second-order** $(\delta t)^2$ (two variables in one bin), the main term **first-order** δt (one variable per bin). Refining the bins **confirms** the tie vanishes rather than reviving it. Geometrically: two independent continuous random numbers being exactly equal is a probability-zero event.

Hence under atomless marginals,

$$U_A(P, Q) = \sum_{i=1}^n \sum_t F_i^Q(t) \Delta F_i^P(t), \quad \Delta F_i^P(t) = F_i^P(t + \delta t) - F_i^P(t).$$

The most intuitive equivalent form is $U_A = \sum_i \mathbb{E}_{x_i}[F_i^Q(x_i)]$: A bidding x_i wins with probability "B bids below x_i ," i.e. $F_i^Q(x_i)$; take expectation over x_i for the per-battlefield expected score.

3. Deriving the Value $n/2$

3.1 Lower bound: the first-order condition forces a uniform marginal

Fix P and consider the inner $\min_Q$. The budget $\sum_i y_i = X$ translates to marginals as $\sum_i \mathbb{E}[y_i] = X$, where $\mathbb{E}[y_i] = \sum_t t \Delta F_i^Q(t)$. The Lagrangian:

$$\mathcal{L} = \sum_{i=1}^n \sum_t F_i^Q(t) \Delta F_i^P(t) + \lambda \left(\sum_{i=1}^n \sum_t t \Delta F_i^Q(t) - X \right).$$

Taking the variational derivative w.r.t. $F_i^Q(t)$ (only that point's term contains $F_i^Q(t)$):

$$\frac{\partial \mathcal{L}}{\partial F_i^Q(t)} = \Delta F_i^P(t) - \lambda \delta t.$$

Role dislocation. Setting this to 0 gives $\Delta F_i^P(t) = \lambda \delta t$, i.e. $f_i^P(t) = \lambda = \text{const}$. The equation contains **no** F^Q . Differentiating w.r.t. F^Q yields a condition on f^P —this dislocation says: " P uniform" $\iff$ " U_A is unchanged as B moves Q ," i.e. **B is indifferent among all Q** . This is the signature mechanism of a mixed equilibrium: A's uniform mixing flattens B into "everything is equally good," leaving B no profitable deviation.

$$f_i^P = \text{const} \implies x_i \sim \text{Uniform}[0, c].$$

3.2 Substitute and pin the value with two budget constraints

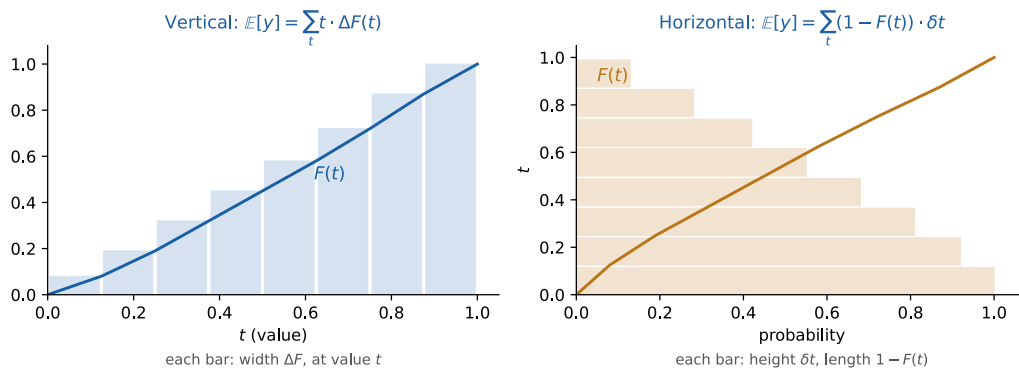


Figure 1: Two ways to slice the mean. Vertical slicing by "value $\times$ probability mass" $\sum_t t \Delta F$; horizontal slicing by "remaining probability per layer $\times$ height" $\sum_t (1 - F) \delta t$. The same area (between the curve and the top edge $F = 1$) equals both. This is the geometric origin of $\mathbb{E}[y_i] = \sum_t (1 - F_i^Q(t)) \delta t$.

When P^* is uniform, $\Delta F_i^P(t) = w$ (independent of t). With interval length c and $k = c/\delta t$ bins summing to mass 1, we get $w = \delta t/c$. Substituting and using $\mathbb{E}[y_i] = \sum_t (1 - F_i^Q(t))\delta t$ (horizontal slicing of the mean) gives $\sum_t F_i^Q(t)\delta t = c - \mathbb{E}[y_i]$:

$$U_A(P^*, Q) = \frac{1}{c} \sum_{i=1}^n (c - \mathbb{E}[y_i]) = n - \frac{1}{c} \sum_{i=1}^n \mathbb{E}[y_i] = n - \frac{X}{c}.$$

This **contains no** Q -equal for every Q , precisely indifference. Now pin c using two budgets:

1. P^* uniform $\implies \mathbb{E}[x_i] = \frac{c}{2}$ (uniform mean = interval midpoint).
2. A obeys the budget $\implies \sum_i \mathbb{E}[x_i] = X$.

Combining: $\frac{c}{2}n = X \implies c = \frac{2X}{n}$. Therefore

$$U_A(P^*, Q) = n - \frac{X}{2X/n} = n - \frac{n}{2} = \frac{n}{2} \quad \forall Q.$$

A conclusion, not an assumption. We presupposed **no** X , F , or $n/2$. The first-order condition forces uniformity; substitution plus two budgets pins $c = \frac{2X}{n}$, and $\frac{n}{2}$ falls out on its own. Incidentally, the CDF of $x_i \sim \text{Uniform}[0, \frac{2X}{n}]$ is exactly $F(t) = \frac{t}{c} = \frac{nt}{2X}$ —the marginal so often handed to readers as an "assumption" is here a **forward** conclusion. **Easter egg:** the cap $\frac{2X}{n}$ is exactly twice the even split $\frac{X}{n}$.

Hence $g(P^*) = \min_Q U_A(P^*, Q) = \frac{n}{2}$, giving the lower bound $\underline{v} \geq \frac{n}{2}$.

3.3 Upper bound: a designated symmetric Q_0 dominates any P

For any P , $g(P) = \min_Q U_A(P, Q) \leq U_A(P, Q_0)$. Take Q_0 uniform ($F_i^{Q_0}(t) = t/c$):

$$U_A(P, Q_0) = \frac{1}{c} \sum_{i=1}^n \underbrace{\sum_t t \Delta F_i^P(t)}_{= \mathbb{E}[x_i]} = \frac{X}{c} = \frac{n}{2}.$$

So for **any** P : $g(P) \leq \frac{n}{2} \implies \underline{v} = \max_P g(P) \leq \frac{n}{2}$. With the lower bound: $\underline{v} = \frac{n}{2}$.

3.4 The symmetric minimax side

Weak duality gives only an inequality, so an independent argument is needed. **Upper** $\bar{v} \leq \frac{n}{2}$: since $U_A(P, Q) = \frac{n}{2} \quad \forall P$, taking $\max_P$ is still $\leq \frac{n}{2}$. **Lower** $\bar{v} \geq \frac{n}{2}$: for any Q , $\max_P U_A(P, Q) \geq U_A(P^*, Q) = \frac{n}{2}$; take $\min_Q$, preserving the sign. Combining:

$$\underline{v} = \bar{v} = \frac{n}{2}.$$

Theorem 1 (Value). The continuous Colonel Blotto game has a value, $\underline{v} = \bar{v} = n/2$.

4. Nash Equilibrium

Theorem 2. Let P^*, Q^* be any legitimate strategies whose marginals are $\text{Uniform}[0, \frac{2X}{n}]$. Then (P^*, Q^*) is a Nash equilibrium with value $n/2$.

Nash equilibrium requires neither player can profit by unilateral deviation, i.e. two inequalities:

- **A's side** ($\max U_A$): $U_A(P, Q^*) \leq \frac{n}{2} \forall P$. The upper-bound step gives $U_A(P, Q^*) = \frac{n}{2} \forall P$; equality satisfies $\leq$.
- **B's side** ($\max U_B = n - U_A$): $U_A(P^*, Q) \geq \frac{n}{2} \forall Q$. Indifference gives $U_A(P^*, Q) = \frac{n}{2} \forall Q$; equality satisfies $\geq$.

Both deviation conditions are met for free by the stronger " $= \frac{n}{2}$ " fact. ■

Two layers, don't conflate. $\underline{v} = \bar{v} = \frac{n}{2}$ is **value existence** (a global quantity with both players moving; a premise); $U_A(P, Q^*) \leq \frac{n}{2}$ and $U_A(P^*, Q) \geq \frac{n}{2}$ are **Nash equilibrium** (pin the opponent, move only yourself; a conclusion). In zero-sum games the optimal-strategy sets are **interchangeable**: $P^* = Q^*$ is merely a convenient special case under symmetry, not a requirement.

5. Existence Construction of the Optimal Strategy P^*

So far P^* carries only a **marginal** constraint. But uniform marginals do **not** imply the existence of a legitimate joint distribution with all marginals uniform and $\sum_i x_i = X$ at every draw. It must be constructed.

5.1 A necessary check and two failed attempts

Mean check: uniform marginals $\implies \mathbb{E}[x_i] = \frac{X}{n}$, $\sum_i \mathbb{E}[x_i] = X \checkmark$. But this is only the mean; $\sum_i x_i = X$ demands **every realization** equal X .

Two naive attempts both fail (empirically verified).

- **Full chained shift** ($x_{i+1} = x_i \pm \frac{X}{n}$): marginals uniform $\checkmark$, but $\sum_i x_i$ drifts $\times$.
- **Chain + last-coordinate buffer** ($x_n = X - \sum_{<n}$): budget locked $\checkmark$, but for $n \geq 4$ the coordinate x_n goes out of range (at $n = 6$, a 60% out-of-range fraction) and turns bell-shaped $\times$. Root cause: sums of independent uniforms tend to a bell curve (Irwin–Hall); all residual error is dumped onto a single x_n .

Lesson: the budget constraint must be **spread across all coordinates**, not shouldered by one.

5.2 Pairwise construction (even n)

Proposition 2. Let n be even, $c = \frac{2X}{n}$. Pair the coordinates; for the k -th pair draw $u_k \sim \text{Uniform}[0, c]$ and set $x_{2k-1} = u_k$, $x_{2k} = c - u_k$. Then every marginal is $\text{Uniform}[0, c]$, $\sum_i x_i = X$ holds identically, with no out-of-range values.

Each pair sums to $c = \frac{2X}{n}$; with $\frac{n}{2}$ pairs the total is $\frac{n}{2} \cdot \frac{2X}{n} = X$ (each pair self-locks). u_k uniform $\implies c - u_k$ reflected is still uniform. $u_k \in [0, c] \implies$ both coordinates lie in $[0, c]$. ■

Key: the budget is **spread across each pair**, and error is absorbed in-pair, never accumulating.

5.3 Pairs + a triple (odd n)

Odd n splits into $\frac{n-3}{2}$ pairs plus one triple (summing to **rem** $= \frac{3c}{2}$). The triple uses the single-parameter $n = 3$ construction: draw $U \sim \text{Uniform}[0, c]$,

$$x_a = U, \quad x_b = \begin{cases} U + \frac{c}{2} & U < \frac{c}{2} \\ U - \frac{c}{2} & U \geq \frac{c}{2} \end{cases}, \quad x_c = \frac{3c}{2} - x_a - x_b.$$

All three marginals are $\text{Uniform}[0, c]$ (the two linear pieces of x_c cover the same interval, and their densities superpose to the constant $2/c$), and the sum is identically $\frac{3c}{2}$. Scale is self-consistent: the triple's native sum $\frac{3c}{2}$ equals exactly the remainder.

Concrete instance n=7 ($X = 1, c = \frac{2}{7}$). Two pairs + one triple: $x_1 = u_1, x_2 = \frac{2}{7} - u_1; x_3 = u_2, x_4 = \frac{2}{7} - u_2; x_5 = U, x_6 = U \pm \frac{1}{7}, x_7 = \frac{3}{7} - x_5 - x_6$. Total $\frac{2}{7} + \frac{2}{7} + \frac{3}{7} = 1$ ✓. A real sample: $[0.063, 0.223, 0.234, 0.052, 0.144, 0.001, 0.284]$, sum = 1, each entry $\leq \frac{2}{7} \approx 0.286$.

Theorem 3 (Existence). For every $n \geq 2$ there exists a legitimate joint $P^* \in \Delta(S_X)$ with each marginal $\text{Uniform}[0, \frac{2X}{n}]$ and $\sum_i x_i = X$ almost surely. Hence the equilibrium strategy P^* exists.

Even n by the pairwise construction; odd n by $\frac{n-3}{2}$ pairs plus a triple. Both were verified by 4×10^6 Monte-Carlo draws: budget identically = X , zero out-of-range, per-coordinate 0.25/0.5/0.75 quantile deviation $\sim 10^{-4}$. ■

5.4 Per-n verification (Monte Carlo, 4×10^6 draws each)

The tables contrast the **correct construction** with the **failed buffer method**. The correct construction, for all n , keeps budget = X , zero out-of-range, marginal quantile deviation $\sim 10^{-4}$ (sampling noise); the buffer method is lucky at $n = 2, 3$ and, at $n \geq 4$, goes out of range with a bell-shaped last coordinate.

n	method	budget $\backslash = X ?$	out-of-range	max quantile dev.
2	pairs	✓	0.00%	0.0002
3	triple	✓	0.00%	0.0004
4	pairs	✓	0.00%	0.0003
5	pairs+triple	✓	0.00%	0.0003
6	pairs	✓	0.00%	0.0003
7	pairs+triple	✓	0.00%	0.0006
8	pairs	✓	0.00%	0.0003

Failed buffer method (chained shift + last-coordinate buffer), behavior of x_n :

n	budget $\backslash = X ?$	out-of-range	$\backslash x_n$ quantile dev.	verdict
2	✓	0.00%	0.0002	lucky
3	✓	0.00%	0.0004	lucky
4	locked	8.33%	0.0838	fails
5	locked	10.0%	0.1253	fails
6	locked	10.0%	0.1502	fails
7	locked	9.52%	0.1671	fails
8	locked	8.93%	0.1789	fails

The out-of-range fraction stabilizes near **9%**, and the x_n quantile deviation climbs from 10^{-4} to **0.18** (at $n = 8$, uniformity is nearly gone)—exactly the Irwin–Hall bell-curve effect, with all residual dumped on one coordinate.

6. Global Review

The complete derivation chain (all forward, no reverse-engineering):

$$\begin{array}{l}
 \max_P \min_Q U_A \xrightarrow{\text{1st-order}} P^* \text{ uniform marginal} \xrightarrow{\text{subst. + 2 budgets}} c = \frac{2X}{n}, \text{ value} = \frac{n}{2} \\
 \xrightarrow{\text{any } P \text{ upper bd}} \underline{v} \leq \frac{n}{2} \xrightarrow{\text{sym. minimax}} \underline{v} = \bar{v} = \frac{n}{2} \xrightarrow{\text{deviation}} \text{NE} \xrightarrow{\text{pairs/triple}} P^* \text{ exists.}
 \end{array}$$

The marginal $F(t) = \frac{nt}{2X}$ is a **conclusion**, not an assumption.

Methodological notes

1. **Define the object first, presuppose no solution.** Everything proceeds from $\max_P \min_Q U_A$; the value, optimal marginal, and symmetric equilibrium are all **derived out**.

2. **The extremum of a linear objective is on the boundary.** U_A is linear in P and in Q ; "interior stationary points" are meaningless. Optimality is characterized by indifference (opponent flattened) and endpoint comparison, not by a bare derivative set to zero.
3. **For constructive claims, code is the final arbiter.** Whether marginals are uniform, whether the budget holds, whether values go out of range—geometric intuition errs easily here (several intuitive misjudgments during this derivation were corrected by Monte Carlo). Correlation $\neq$ marginal shape: a joint with highly correlated coordinates can still have every marginal uniform.

Appendix A: Derivation Log / Key Exchanges

This section faithfully records the sticking points and corrections during the derivation—this is a research note, not a clean paper, and the trial-and-error and self-correction are themselves part of the content. Written in first person ("I" refers to the AI acting as derivation companion, whose judgments erred at several points and were corrected by the human collaborator or by code).

A.1 The $\sum_t$ notation dispute. I once assumed $\sum_t$ meant "sum over discrete atoms," and on that basis judged $\sum_t \Pr(x_i = t)(\dots) = 0$ in the continuous case. The collaborator objected: "a sum is just notation, it need not be discrete." Correct resolution: $\sum_t$ should be read as an integral against the measure dF (Stieltjes), with weight the **infinitesimal measure** $dF(t) = f(t)dt$ (nonzero in the continuous case), not the **single-point probability** $\Pr(x_i = t)$ (zero in the continuous case). I had written $dF^P(t)$ as $\Pr(x_i = t)$, yielding the absurd $U_A \equiv 0$, caught on the spot. Lesson: in the continuous regime the bare sum $\sum_t F(t)$ is pathological ($\rightarrow \infty$); it must be paired with δt to form an area.

A.2 "The assumption is derived, not assumed." Midway, the collaborator halted the derivation: textbooks often hand readers $F(t) = \frac{nt}{2X}$ as an assumption, which is reverse-engineering (working backward from a known answer). The demand: **demote it to a conclusion**—presuppose no P^* , no F , no $n/2$, and grow everything forward from the $\max_P \min_Q U_A$ first-order condition. This was the defining moment for the whole first-principles stance.

A.3 The inequality sign, flipped repeatedly. Proving the maximin upper bound, the collaborator once wrote $g(P) \geq g(P^*)$, i.e. "every P 's guaranteed value is $\geq P^*$'s"—which would make P^* A's **worst** choice, contradicting maximin's goal of maximizing g . Correction: it should be $g(P) \leq g(P^*)$. A purely algebraic anchor— U_A is a weighted sum of F^Q with weights ΔF^P ; if P is uniform the weights are equal everywhere and B has nowhere to hide; if P is non-uniform the weights vary and B targets the light ones to push U_A lower. So non-uniformity "hands B a gift," $g(P_{\text{non-unif}}) < g(P^*)$.

A.4 Three intuitive misjudgments corrected by code (construction).

- **Misjudging GPT's $n = 3$ construction.** I claimed the single-parameter $n = 3$ construction had a non-uniform x_3 marginal, on the faulty intuition " x_1, x_2 strongly correlated $\implies$ marginal must skew." Recomputing, x_3 's density is $\frac{3}{2}$ (the two linear pieces cover the same interval and superpose to a constant), exactly uniform. I was wrong. **Correlation $\neq$ marginal shape**: a highly correlated joint can still have every marginal uniform.
- **Cold water on the buffer idea.** Against the collaborator's buffer idea I first ran the empirics—this cold water was correct ($n \geq 4$ out-of-range + bell shape), but I should not have used a "judging" tone right after crashing myself; run the code first.
- **Odd- n scale misjudgment.** I worried about a scale conflict ("the triple can only make $\frac{X}{5}$ yet a coordinate must reach $\frac{2X}{5}$ "), shouting it out on feel. Computing rem: for $n = 5$, $\text{rem} = 1 - \frac{2}{5} = \frac{3}{5}$, exactly the triple's native sum $\frac{3c}{2} = \frac{3}{5}$; the scale is self-consistent. I concluded before computing rem, and erred again.

Meta-lesson: for constructive claims, geometric intuition errs easily and code (Monte Carlo) is the sole arbiter. The human collaborator drove out all correct constructions; the AI's three intuitive misjudgments were each corrected by empirics.

A.5 The tie term's δt tug-of-war. The collaborator repeatedly pressed "where does the tie go in the continuous case," from a worry about "point-to-point equality remaining inside a bin" to the final confirmation: as $\delta t \rightarrow 0$ the same-bin probability $\sim \delta t \rightarrow 0$ squeezes the tie from above, and the equal-points set inside a bin is **nonempty but measure-zero**. Refining bins **confirms** the vanishing rather than reviving it. This was the intuition point polished longest in the whole note.